Decision Models on Bengali: Accuracy, Calibration and Script Robustness under Rule-Based and AI-Generated Romanization

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Abstract

Decision models answer a typed question in one forward pass and return a probability for every option. I tested one hosted decision model, Jev, ten open decision models and one general-purpose language model on five Bengali tasks, zero-shot, in Bengali script and in Latin letters. Jev scored highest on all five tasks among the models run on the full test sets; on Belebele it reached 0.918 (95% CI 0.897 to 0.936). The open d3-flash matched or beat it on three tasks, on a subset of about 405 items per task. Rule-based romanization lowered every model's accuracy on every test set where it was run. On 877 items that ChatGPT romanized, Grok and Claude corrected and I reviewed (AI-generated Banglish), it lowered accuracy less than rule-based romanization for every model but Julia-1, and for d3-edge, Laya-multilingual and Julia-1 the loss against Bengali is within noise. Jev lost 11.2 points against the Bengali original, where rule-based romanization of the same text cost it 19.4. The model ranking changed little between the two (Kendall $\tau = 0.89$ over 11 models on all 877 items from SentNoB, MASSIVE and XNLI-bn; 0.78 on the 383 SentNoB and XNLI-bn items that all twelve systems answered), so the size of a romanization penalty depends on how the Latin-script text is made more than the ordering of models does. None of this text was typed by a person, so how it compares with real Banglish typing is untested. The difference is concentrated in MASSIVE; on SentNoB, the only native text, it is at most 7.1 points for any model. Every open model was overconfident on Bengali text, and one temperature fitted on a separate development set removed most of that, but less of it on rule-based romanized text. Code, tables and the AI-generated Banglish are at `https://github.com/ihpolash/system-one-testing-for-bengali`.

1 Introduction

A decision model reads a text and a typed question and returns one probability per option, without generating anything. That makes the output easy to act on: a pipeline can accept confident answers and send the rest to a larger model or a person [4, 16]. Nearly all of the evidence for this comes from English. The one study that left English shows why that matters. On Chinese tasks Laya-multilingual scored 40.2% where Jev scored 68.4% [19]. For Bengali I found only the per-language Belebele and SIB-200 numbers that ship with the Jev benchmark [4]. I found no study that asked whether a decision model's confidence still means something in Bengali.

Bengali has a second difficulty. A great deal of everyday Bengali is typed in Latin letters, with no standard spelling. Work on other Indic languages and on Bangla dialects tests romanized text produced by a transliteration tool, and finds large losses [11, 12]. A tool applies one rule per letter.

Study	Decision models	Bengali	Romanized	Calibration	Human-typed text
Deußer et al. [4]	Jev only	per-language file	Belebele, SIB-200	yes	no
Wang et al. [19]	trained model	no	no	yes	no
Khullar et al. [12]	no	no	tool-made	routing only	no
Jawad et al. [11]	no	dialects	tool-made	no	no
This paper	12 systems	5 tasks	rule, natural-style, AI-generated	yes	no (877 AI-generated items)

Table 1: What each closest study covers, from reading the papers. A "no" means the paper does not report it.

People do not, and a language model writes Banglish in yet another way. I found no measurement, for Bengali, of how much of a model's loss on tool-made romanization carries over to Latin-script text produced by other means.

I measured it against one alternative: Banglish generated by ChatGPT, corrected with Grok and Claude, and reviewed by me. I scored all models on 935 sentences from three datasets in four spellings, with that text as one of them (Section 4.2). Around it sits a zero-shot benchmark of twelve systems on five tasks, with calibration, risk–coverage and option-order analyses (Section 4). Every number here is produced by the released code, and the harness reproduces published numbers from independent groups for Jev, Laya-multilingual and laya-bangla (Section 4.5). This is not the first Bengali result for any decision model, since Jev's Belebele and SIB-200 numbers exist. Kev-9B, Kev-27B, d3-27B and Microsoft-Decision-1 are not covered.

2 Related work

Table 1 shows what the closest studies cover. In English, decision models have been checked for calibration [16], for annotation work with open models [9], for sensitivity to option names [18] and to option order [13], and for cost and wording robustness [17]. Other languages are thin. Chinese-Jev trains a Chinese model [19], and one paper compares Jev's cultural-value answers in English and Arabic [2]. Script robustness has been studied for generative models only [11, 12], and the Bengali half of the large romanization corpus of Gharami et al. [7] is generated by rule, not typed by people. For calibration and selective prediction I use the standard measures [6, 8]; the option-rotation test follows Zheng et al. [20]. The tasks are Belebele [3], SIB-200 [1], MASSIVE [5], SentNoB [10] and the Bengali translation of XNLI.

3 Benchmark

Models. Jev (`jev-1.13.0`) is reached through its API. The open models ran on Kaggle GPUs: Laya-multilingual, an mmBERT-based encoder [15]; laya-bangla v2, a Bangla fine-tune of it (commit `4c5eb48932`); Julia-1, with 144M parameters; the vLLM-SR Decision 3.0 models d3-edge, d3-lite, d3-nano, d3-mini and d3-flash [14]; and Kev-0.8B and Kev-4B (release 1.0, tag `v1.0`), served with the project's own code in bf16 with its shipped temperature. Kev's model cards list English as its only language and name other languages as a known weakness. Kev-4B sits at the memory limit of a 15 GB T4, so for it I switched off CUDA-graph replay and the prefix cache, which change speed and memory but not the model, after a first run ran out of memory. Kev was scored on the test sets and on the AI-generated Banglish items only, so it is missing from the calibration and option-order results. d3-flash has 9B parameters and did not fit comfortably, so it ran 4-bit (NF4) on a hash-ordered subset of about 405 items per task. Gemma-4-E4B-it is not a decision model; it serves as a baseline, with each option scored by the next-token probability of its label

Dataset	Task	Options	Metric	Text origin	Test items (all / used)
Belebele	reading comprehension	4 options	accuracy	human translation (FLORES-200)	900 / 900
SIB-200	topic	7 options	accuracy	human translation (FLORES-200)	204 / 204
XNLI-bn	natural language inference	3 options	accuracy	machine translation of XNLI	4,895 / 1,500
MASSIVE bn-BD	intent	60 options	accuracy	human translation	2,974 / 1,500
SentNoB	sentiment	3 options	macro-F1	native social-media comments	1,586 / 1,586

Table 2: Tasks. XNLI-bn is the machine-translated Bengali XNLI, so its text is translationese; SentNoB is the only native Bengali set. MASSIVE is posed with all 60 intents; the test items use 59 of them. The SentNoB split is a third-party mirror whose provenance we could not verify.

and the probabilities normalized over the options. Kev-9B and Kev-27B do not fit on that GPU, and d3-27B and Microsoft-Decision-1 (API only) were not run.

Tasks. The five tasks are in Table 2. Each item is a single-choice question with English instructions and option names; only the passage or utterance changes between conditions. The exact question structure for every task is in `docs/PROMPTS.md` in the repository. I cut XNLI-bn and MASSIVE to hash-ordered subsets of 1,500 items so that any smaller run is nested in a larger one. Bengali option names changed Jev's accuracy by at most 0.8 points on SentNoB, XNLI-bn and SIB-200, so I kept English names throughout. The metric is accuracy, and macro-F1 for SentNoB.

Text conditions. *Bengali* is the dataset text. *Rule-based romanization* maps each letter to Latin letters by a fixed rule set, without diacritics. *Natural-style romanization* replaces each word with its most frequent spelling in a public corpus of natural Banglish (nahidstaq/bangla-transliteration-data) and falls back to rules tuned on held-out words from that corpus. Both are synthetic. Belebele passages were not romanized.

AI-generated Banglish. I took 935 Bengali sentences from SentNoB, MASSIVE and XNLI-bn; spelling errors in 81 of them were corrected before romanization. ChatGPT (Luna 6.1) generated a Banglish rendering of every sentence. Grok (4.7) and Claude (Sonnet 5.5) were then used to correct those renderings, and I supervised and reviewed the corrections; about 35% of the renderings were changed. I corrected the spelling of the Bengali source myself. No sentence was typed by a person from scratch, so this condition is AI-generated, author-reviewed romanization, not a sample of how people type, and I call it AI-generated Banglish throughout. I did not record the prompts. The repository holds a reference prompt of the same kind (`docs/BANGLISH_GENERATION.md`), written afterwards and not the one used. Items are matched to the datasets by their original text. An XNLI-bn item needs both sentences, so the usable set is 877 items: 494 from MASSIVE, 325 from SentNoB and 58 XNLI-bn pairs. No sentence was dropped; Table 3 shows how 935 sentences become 877 items. Spelling correction alone moved Jev by +0.2 points (95% CI -0.2 to +0.7) and no model by more than 0.6, so I compare the AI-generated Banglish with the corrected Bengali, and I apply both romanizers to the corrected text as well.

Statistics. Intervals are 95% percentile bootstraps (seed 0). On the test sets they resample premise or passage clusters (Belebele, XNLI-bn) or items, 1,000 times (400 for the SentNoB macro-F1, 500 for paired differences). On the 877 AI-generated Banglish items they resample items, 1,000 times, and every comparison between conditions is a paired difference on the same items. Calibration is measured with expected calibration error over 15 equal-width bins (ECE), the signed

Dataset	Sentences romanized	of which spelling corrected	Scored items
MASSIVE	494	8	494 (utterance)
SentNoB	325	70	325 (comment)
XNLI-bn	116	3	58 (premise–hypothesis pair)
Total	935	81	877

Table 3: From sentences to scored items. An XNLI-bn item is a premise–hypothesis pair, so two sentences make one item; nothing was excluded.

gap between confidence and accuracy, the Brier score and risk–coverage curves. I report two ECE aggregations and they differ on purpose. The primary one is computed within each task on the full test sets and then averaged without weights over XNLI-bn, MASSIVE and SentNoB (Table 7; per task in Table 11); I prefer it because the tasks differ in the number of options and in label balance, so no single task should set the value. The other pools the 877 items across tasks (Table 12), so MASSIVE weighs 56% of it; it is the only calibration measure available for the AI-generated Banglish, and the two should not be compared number for number. Temperature scaling uses one scalar per model and task. It minimizes the mean negative log-likelihood over a grid of 41 log-spaced temperatures from 0.2 to 6, on a development split of 1,500 items (XNLI-bn, MASSIVE, SentNoB) that is disjoint from the test items and holds Bengali text only. The fitted temperature is then applied unchanged to the test items in Bengali and in rule-based romanization; no test label enters the fit. Belebele and SIB-200 have no development split, and d3-flash and Kev were not run on one, so they have no scaled results. The confidence gate uses the same development split: its threshold is the smallest one whose accepted dev accuracy reaches 90%, with at least 5% coverage. Option-order sensitivity uses three cyclic rotations of the options. I report no significance tests. The main comparison is AI-generated Banglish against rule-based romanization on the pooled items; every other comparison is descriptive, and none is corrected for the number of comparisons.

4 Results

4.1 Accuracy on Bengali script and romanized text

Table 4 and Figure 1 give the main results. Jev has the highest point estimate on every task among the models run on the full sets. The closest open model is d3-nano on XNLI-bn (0.746 against 0.751), with d3-mini at 0.729, and the intervals in Table 9 overlap. d3-flash, on its subset, scores 0.828 on XNLI-bn against Jev's 0.752, and trails on Belebele (Table 5). It is above Jev on three tasks, but with 204 to 405 items per task I read that as level. Kev-4B lands next to d3-lite: 0.561 on Belebele against 0.577, 0.779 on SIB-200 against 0.760, 0.536 on SentNoB against 0.535, and below it on XNLI-bn (0.624 against 0.704) and MASSIVE (0.575 against 0.624). Kev-0.8B scores 0.632 on SIB-200 and 0.372 on MASSIVE, and falls in the lower group with d3-edge and Laya-multilingual. Both Kev models are well below Jev on every task. Gemma-4-E4B holds up best under romanization: on SIB-200 it keeps 0.730 of 0.853, where d3-mini keeps 0.549 of 0.838. laya-bangla beats its base model on SIB-200 (0.750 against 0.588) and on MASSIVE (0.575 against 0.179) and still sits below Jev, d3-mini and Gemma-4-E4B on every task. On SentNoB, always answering with the majority class scores a macro-F1 of 0.195 and uniform random guessing 0.328. Laya-multilingual reaches only 0.263, below random guessing, Kev-0.8B 0.359 and Julia-1 0.403. Julia-1 cannot take the 60-option MASSIVE menu at all. Laya-multilingual's SentNoB score also depends on the wording of the question: macro-F1 is 0.263 with my default instruction, 0.321 with

Model	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.918	0.897 (0.725)	0.751 (0.572)	0.757 (0.481)	0.623 (0.527)
Gemma-4-E4B	0.773	0.853 (0.730)	0.631 (0.519)	0.689 (0.429)	0.590 (0.533)
d3-mini	0.727	0.838 (0.549)	0.729 (0.562)	0.749 (0.383)	0.579 (0.486)
d3-nano	0.702	0.848 (0.377)	0.746 (0.497)	0.705 (0.221)	0.592 (0.408)
Kev-4B	0.561	0.779 (0.451)	0.624 (0.475)	0.575 (0.226)	0.536 (0.408)
d3-lite	0.577	0.760 (0.265)	0.704 (0.463)	0.624 (0.121)	0.535 (0.343)
laya-bangla	0.313	0.750 (0.250)	0.625 (0.435)	0.575 (0.077)	0.464 (0.216)
d3-edge	0.409	0.475 (0.235)	0.523 (0.445)	0.243 (0.044)	0.472 (0.366)
Kev-0.8B	0.374	0.632 (0.265)	0.469 (0.413)	0.372 (0.118)	0.359 (0.261)
Laya-multilingual	0.287	0.588 (0.275)	0.635 (0.383)	0.179 (0.045)	0.263 (0.141)
Julia-1	0.287	0.520 (0.284)	0.382 (0.316)	–	0.403 (0.299)
Majority class	–	0.250	0.342	0.071	0.195
Uniform random	0.250	0.143	0.333	0.017	0.328

Table 4: Accuracy on Bengali script, and in brackets on rule-based romanization, rotation 0 (SentNoB: macro-F1). Belebele passages were not romanized. Julia-1 cannot take the 60 options of MASSIVE. Majority class: the share of the most frequent label in the test items (for SentNoB, the macro-F1 of always predicting it); Belebele's answer positions rotate, so it has none. Uniform random: one over the number of options (for SentNoB, the expected macro-F1 of random guessing). Intervals are in Table 9.

On the d3-flash subset	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.840	0.897 (0.725)	0.752 (0.557)	0.799 (0.519)	0.637 (0.577)
d3-flash (4-bit)	0.730	0.892 (0.627)	0.828 (0.598)	0.819 (0.412)	0.653 (0.519)

Table 5: d3-flash (4-bit) and Jev on the same hash-ordered subset (204 to 405 items per task); romanized in brackets. d3-flash is the 9B checkpoint in 4-bit quantization.

option names only, and 0.376 with Bengali wording. I report the default and flag the range.

4.2 AI-generated Banglish

Figure 2 and Table 6 put four spellings of the same 877 items side by side. The pooled figures are plain accuracy over three tasks with different numbers of options, so they describe this item set and are not a cross-task quality score; the per-task comparisons in Table 8 are the primary evidence. Every model but Julia-1 does better on the AI-generated Banglish than on rule-based romanization, by 8.2 to 17.1 points. For Jev the gain is +8.2 points (95% CI +5.1 to +11.1). Natural-style romanization lands between the two, so tuning a romanizer on typed Banglish narrows the gap but does not close it. The gain varies by task (Figure 3). For Jev it is +13.6 points on MASSIVE (+9.7 to +17.4), +8.6 on XNLI-bn (+0.0 to +17.2, from only 58 pairs), and +0.0 on SentNoB (-4.9 to +4.6). Because MASSIVE makes up 56% of the pooled items, Table 8 gives each task separately and the unweighted mean of the three task differences. For Jev that mean is 7.4 points, and across the models other than Julia-1 it runs from 6.4 to 14.3, so the gain does not come from the weighting. It does come mostly from MASSIVE. On SentNoB the interval for the gain includes zero for six of the twelve systems, and for Julia-1 it is negative. My guess is that MASSIVE's short commands, full of English loan words, are what a rule-based tool gets wrong and a language model spells in the conventional way. I did not test that. Julia-1, near chance on these items, shows no gain.

The order of models moves little, but how little depends on the items. Over the 11 models that answered all 877 items, from SentNoB, MASSIVE and XNLI-bn (every system but Julia-

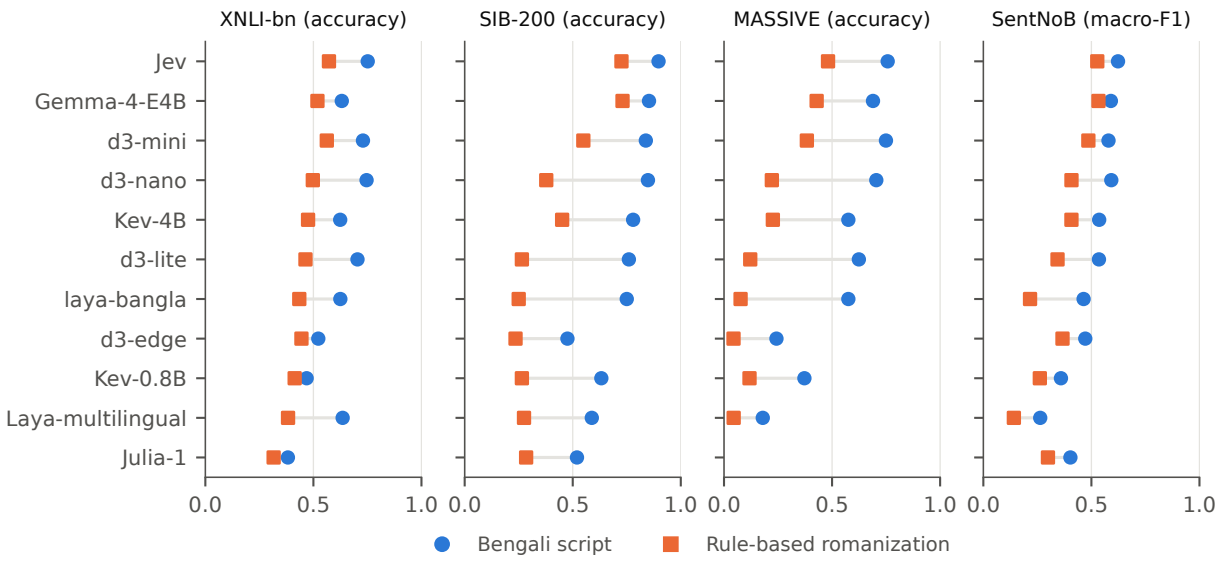


Figure 1: Bengali script against rule-based romanization for each model and task. Lines join the two conditions for the same model.

1), Kendall τ between the ordering on AI-generated Banglish and on rule-based romanization is 0.89 (95% bootstrap interval 0.75 to 0.93), and 0.93 (0.82 to 1.00) for natural-style. On the 383 SentNoB and XNLI-bn items that all 12 systems answered, which leaves out MASSIVE because Julia-1 cannot take it, the values are 0.78 (0.60 to 0.88) and 0.84 (0.67 to 0.93). Neither value is a general measure of ranking stability: they describe two different task mixtures. The agreement is clearly positive and clearly short of identical. The systems are scored on the same items, so this is one comparison, not eleven independent ones.

4.3 Calibration and selective prediction

Overconfidence differs a lot between models. Raw ECE on Bengali script, averaged over the three tasks as described above, runs from 0.074 for d3-mini to 0.432 for Julia-1 (Table 7). One temperature, fitted on a separate Bengali development split, lowers ECE for every model on Bengali text; for Julia-1 on XNLI-bn it takes ECE from 0.420 to 0.098. On rule-based romanized text the same temperature leaves a higher ECE for eight of the nine models with a development run (Kev and d3-flash have none). d3-mini is the exception. Jev's scaled ECE is 0.095 on romanized text against 0.047 on Bengali, and laya-bangla's is 0.172 against 0.034. Figure 4 shows the reliability curves and Figure 5 the risk–coverage curves. Confidence still ranks answers within each condition, but rule-based romanization lowers accuracy at every coverage level, including among the most confident answers. Without any scaling, on the 877 items, raw ECE (Table 12) stays within 0.05 across the three conditions for Jev (0.117, 0.109 and 0.101 for Bengali, rule-based and AI-generated Banglish), d3-mini and the two Kev models. For Gemma-4-E4B, d3-nano, d3-lite, laya-bangla, d3-edge and Laya-multilingual it is highest under rule-based romanization, with AI-generated Banglish in between. Julia-1 is near chance and uninformative here. So the miscalibration under romanization is real for most models but not for all, and I did not fit a temperature for the AI-generated condition.

A confidence gate set on Bengali development text to reach 90% accuracy does not carry over. On MASSIVE, Jev accepts 69% of Bengali items (1036 items) at 91% accuracy, but only 23% of

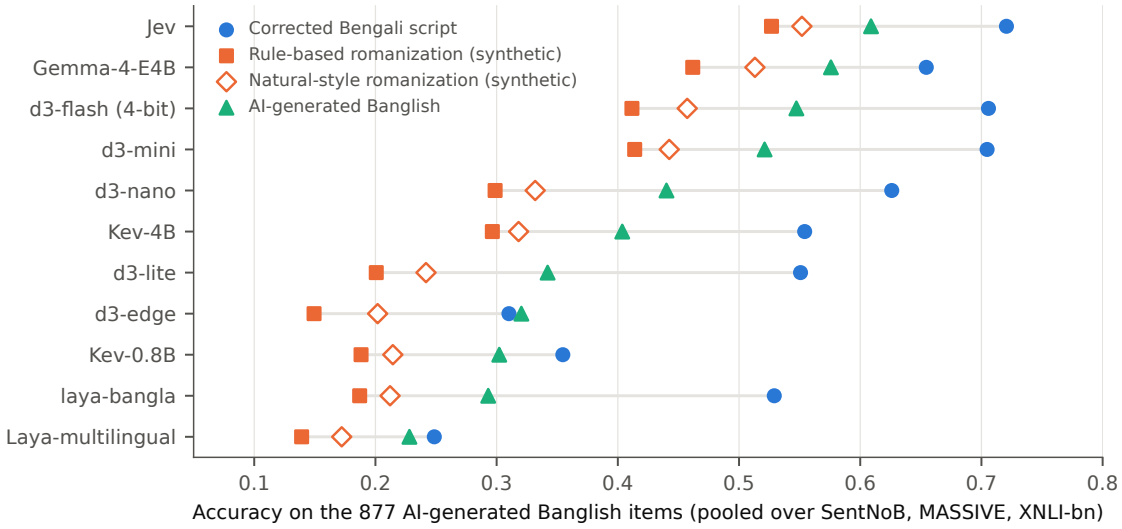


Figure 2: Accuracy of each model on the same 877 items in four spellings.

rule-based romanized items (341), at 88%. d3-mini accepts 68% (1020) at 90% and 22% (335) at 83%. For laya-bangla the accepted rule-based romanized items are 17% accurate, but only 12 items are accepted, so that figure is close to anecdote. So for Jev and d3-mini the gate keeps its accuracy by refusing most rule-based romanized input, and for laya-bangla it fails. This rests on one task and, for some models, on few accepted items, so I treat it as exploratory.

4.4 Option-order sensitivity

Across three rotations of the options, Jev gives the same answer 93% of the time, d3-nano 89%, d3-lite 87% and Gemma-4-E4B 81%. The encoders are far less stable: laya-bangla 67%, Laya-multilingual 53%, Julia-1 35% (Figure 6; per task in Table 10).

4.5 Checks on the harness

Before trusting any of this I tried to reproduce other people's numbers. From the Jev authors' released responses, Belebele ben_Beng accuracy is 0.9167; my own run gives 0.918. For Laya-multilingual I get MASSIVE 0.179 and SIB-200 0.588, against 0.18 and 0.57 to 0.58 reported by the laya-bangla authors. For laya-bangla I get SIB-200 0.750 against their 0.76. Resending 300 Jev requests gave the same answer 99% of the time, and a newer Laya package version changed no answer on 6,563 requests. One comparison I could not make: laya-bangla's authors report MASSIVE 0.81 with a two-step scenario-then-intent protocol, and I ask one 60-way question and get 0.575. The study used 92,618 Jev requests, about 50.9M input tokens and $2.14 at $0.042 per million, plus about 34 GPU hours on Kaggle. I measured latency under mixed conditions, so it is not comparable across models, and I do not report it.

5 Discussion

What a rule-based romanization can and cannot tell you. Rule-based romanization ranks models in nearly the same order as AI-generated Banglish, and costs them more accuracy. For Jev, test data from the rule-based romanizer would suggest losing 19.4 points on Banglish input, about

	Accuracy				Change in points (95% CI)	
Model	Bengali	Rule-based	Natural-style	AI Banglish	AI Banglish − Bengali	AI Banglish − rule-based
Jev	0.721	0.527	0.552	0.609	-11.2 [-14.1, -8.2]	+8.2 [+5.1, +11.1]
d3-flash (4-bit)†	0.706	0.412	0.457	0.547	-15.8 [-19.3, -12.7]	+13.6 [+10.7, +16.8]
Gemma-4-E4B	0.655	0.462	0.513	0.576	-7.9 [-10.7, -5.1]	+11.4 [+8.8, +14.3]
d3-mini	0.705	0.414	0.442	0.521	-18.4 [-22.0, -14.9]	+10.7 [+8.0, +13.6]
d3-nano	0.626	0.299	0.332	0.440	-18.6 [-22.4, -15.1]	+14.1 [+11.3, +17.4]
Kev-4B	0.554	0.296	0.318	0.404	-15.1 [-18.6, -11.3]	+10.7 [+7.5, +13.5]
d3-lite	0.551	0.201	0.242	0.342	-20.9 [-24.2, -17.1]	+14.1 [+11.3, +17.2]
laya-bangla	0.529	0.187	0.212	0.293	-23.6 [-27.3, -20.0]	+10.6 [+8.0, +13.3]
d3-edge	0.310	0.149	0.202	0.320	+1.0 [-2.7, +4.7]	+17.1 [+14.1, +19.7]
Kev-0.8B	0.355	0.188	0.214	0.302	-5.2 [-8.9, -1.7]	+11.4 [+8.4, +14.1]
Laya-multilingual	0.249	0.139	0.172	0.228	-2.1 [-5.0, +0.9]	+8.9 [+6.6, +11.2]
Julia-1‡	0.428	0.441	0.418	0.384	-4.4 [-11.0, +2.1]	-5.7 [-11.2, +0.0]

Table 6: Accuracy on the 877 AI-generated Banglish items (pooled over MASSIVE, SentNoB and XNLI-bn pairs; plain accuracy, SentNoB included) and paired differences with 95% bootstrap intervals over items. †d3-flash is 4-bit. ‡Julia-1 answers only the 383 items it can take.

	Bengali script		Rule-based romanization	
Model	ECE raw	ECE scaled	ECE raw	ECE scaled
Jev	0.126	0.047	0.146	0.095
Gemma-4-E4B	0.303	0.073	0.391	0.082
d3-mini	0.074	0.036	0.079	0.035
d3-nano	0.083	0.041	0.122	0.054
d3-lite	0.076	0.040	0.172	0.108
laya-bangla	0.204	0.034	0.331	0.172
d3-edge	0.174	0.044	0.202	0.085
Laya-multilingual	0.347	0.080	0.530	0.209
Julia-1	0.432	0.104	0.494	0.154

Table 7: ECE (15 bins) on Bengali script and rule-based romanization, before and after one temperature fitted on a native-text development split. Mean over XNLI-bn, MASSIVE and SentNoB.

8 points more than on the AI-generated text. That does not say which of the two is closer to what people type. A language model's Banglish may be cleaner than real typing, and closer to the web text the models were trained on, which would make the loss on real input larger than I measured. What the comparison does show is that the size of a romanization penalty depends on how the Latin-script text is made, so a number measured on one procedure should not be carried over to another. I do not know why the AI-generated text is easier than the rule-based text. Two explanations fit: common words are spelled the way models have seen them, and the spelling corrections removed errors that a tool copies through. The control rules out only the second. Correcting the original Bengali moved no model by more than 0.6 points.

Choosing a model. On Bengali script d3-mini, d3-nano and Gemma-4-E4B come within a few points of Jev on several tasks, so the hosted model's advantage here is modest. Gemma-4-E4B is the most robust open model to romanization, and also among the worst calibrated before scaling (ECE 0.303), which temperature scaling mostly removes. The encoders need care. Their answers change with option order, and Laya-multilingual's SentNoB score moved by more than ten points with the wording of the question.

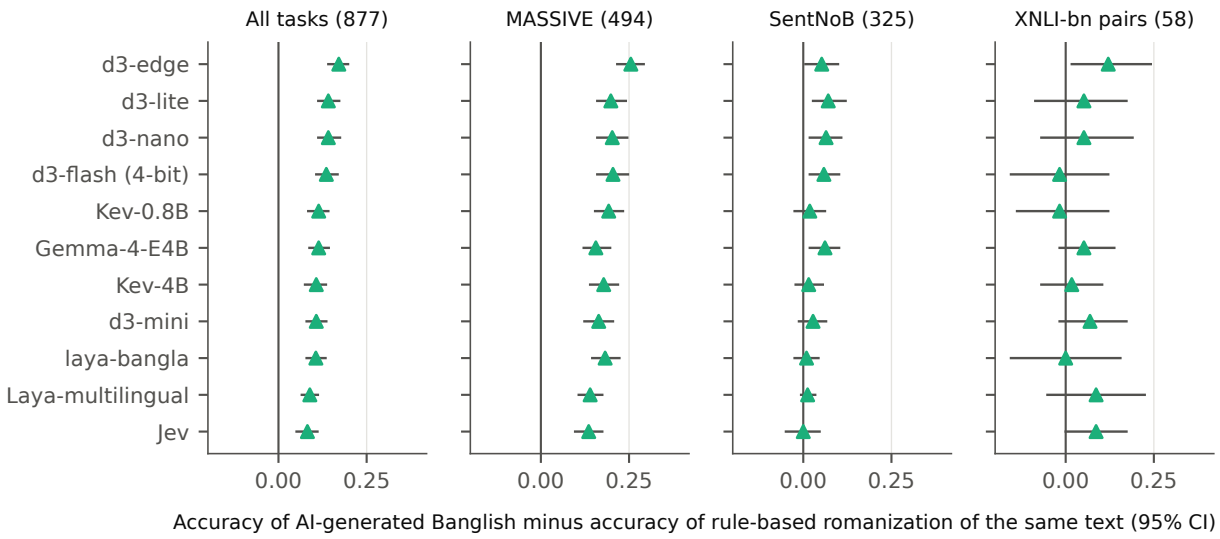


Figure 3: Accuracy of AI-generated Banglish minus rule-based romanization of the same text, pooled and by task. Bars are 95% intervals.

6 Limitations

The Banglish was generated by ChatGPT and corrected with Grok and Claude under my supervision, so this evaluation measures AI-generated, author-reviewed romanization, not text typed by several Bengali speakers. The final spellings may reflect the conventions of those models, and because language-model text may sit close to the data that language models are trained on, it may be easier than real typing for some of the systems I tested. Matching the items controls for item identity; it cannot separate human typing from AI generation. Banglish collected independently from several writers under one protocol is the test of whether these differences generalise, and I have not done that. MASSIVE supplies 56% of the items and XNLI-bn only 58 pairs. XNLI-bn and MASSIVE are translations, so SentNoB is the only native Bengali text, and on SentNoB the AI-generated Banglish is no easier than rule-based romanization for several models. I do not know whether that comes from social-media text or from the generation procedure. d3-flash ran 4-bit on a subset, Kev ran only on the test sets and the AI-generated Banglish items, the larger Kev and d3 models were not run, and Julia-1 cannot take MASSIVE. Contamination is unknown for Jev, Gemma and the Qwen-based d3 models, and laya-bangla was trained on MASSIVE and SIB-200 training splits. The SentNoB split I used is a third-party mirror whose provenance I could not verify. I use one prompt format per model family, and Laya's wording sensitivity shows that this can move a result. The calibration and gate results are exploratory and use no multiplicity correction.

7 Conclusion

On five Bengali tasks Jev leads, several small open models come close on Bengali script, and the open models' overconfidence is largely fixed by one temperature on Bengali text but less fully on rule-based romanized text. Rule-based romanization ranks models in nearly the same order as AI-generated Banglish but costs them more accuracy. A loss measured on one romanization procedure should not be carried to another, and Banglish typed independently by several people is needed to say which procedure matches real use.

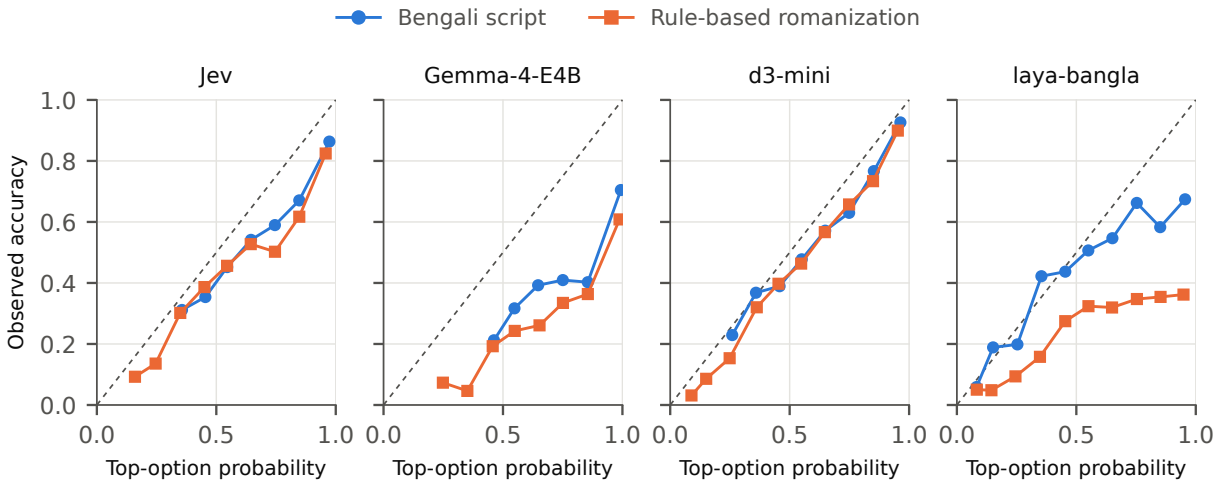


Figure 4: Reliability of four models on MASSIVE, XNLI-bn and SentNoB pooled. The dashed line is perfect calibration.

Availability. The repository is at `https://github.com/ihpolash/system-one-testing-for-bengali`. It holds the harness, the analysis code that generates every table and number in this paper, the summary result tables, item-level predictions with probabilities for the open models, SHA-256 checksums for the released files, the exact model identifiers and settings (`docs/MODELS.md`), the question structure for each task (`docs/PROMPTS.md`), and the AI-generated Banglish for the MASSIVE and XNLI-bn sentences. The predictions contain no dataset text except Belebele's option labels, which are the Bengali option texts (CC BY-SA 4.0). SentNoB is distributed under a no-derivatives licence, so the repository contains neither its text nor the Banglish of its sentences. Jev's raw responses and probability vectors are not included (provider terms); the table for the 877 items holds only Jev's per-item correctness and top-option confidence. The original generation prompts were not recorded; the repository's prompt is retrospective and labelled as such.

Declaration on the use of generative AI

The author used generative AI in two ways. An AI assistant (Claude, Anthropic) supported implementation, analysis and figure scripting, and the drafting and editing of this manuscript. For the Banglish condition, ChatGPT (Luna 6.1, OpenAI) generated a Latin-script rendering of every Bengali sentence, and Grok (4.7, xAI) and Claude (Sonnet 5.5, Anthropic) were used to correct those renderings; the author supervised and reviewed this process. The Banglish is therefore AI-generated, author-reviewed text, as stated in Section 3 and in the limitations. All of this work was carried out under the author's direction and review. The research questions, benchmark design, evaluation protocol and the interpretation of the results are the author's. All numerical results in the tables and text are generated by the released analysis code from the stored results, and every reference was verified by the author against its primary source. The author takes full responsibility for all content of this paper, irrespective of how it was drafted.

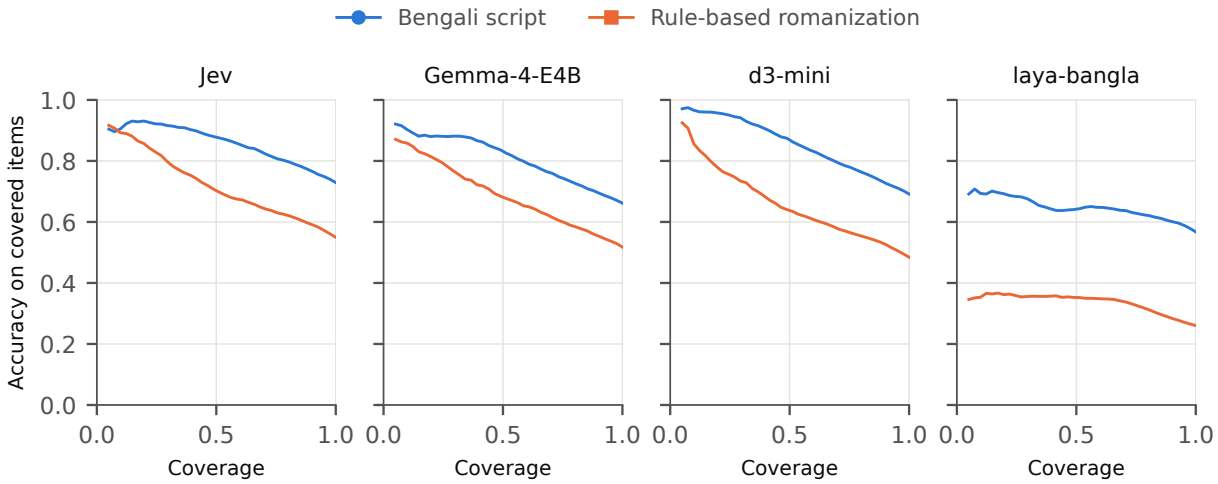


Figure 5: Accuracy on the covered items as coverage grows from the most confident answers.

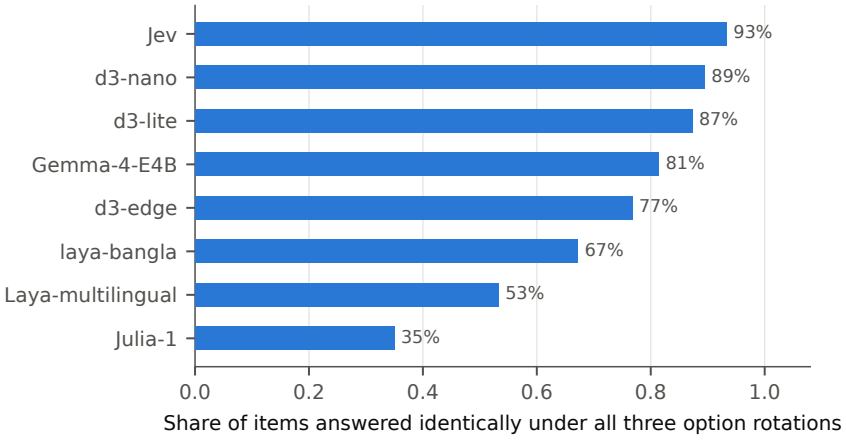


Figure 6: Share of items answered identically under all three option rotations, over all tasks.

References

[1] David Ifeoluwa Adelani, Hannah Liu, Xiaoyu Shen, Nikita Vassilyev, et al. SIB-200: A simple, inclusive, and big evaluation dataset for topic classification in 200+ languages and dialects, 2023.

[2] Bushra Asseri et al. Calibrated to whom? persona and language effects on cultural values in JEV, 2026.

[3] Lucas Bandarkar, Davis Liang, Benjamin Muller, Mikel Artetxe, et al. The Belebele benchmark: a parallel reading comprehension dataset in 122 language variants, 2023.

[4] Tobias Deußer, Lorenz Sparrenberg, and Rafet Sifa. Evaluating and benchmarking the system one model Jev, 2026.

[5] Jack FitzGerald, Christopher Hench, Charith Peris, Scott Mackie, et al. MASSIVE: A 1M-

example multilingual natural language understanding dataset with 51 typologically-diverse languages, 2022.

[6] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017. arXiv:1705.08500.

[7] Kanchon Gharami, Quazi Sarwar Muhtaseem, Deepti Gupta, Lavanya Elluri, and Shafika Showkat Moni. Modeling romanized Hindi and Bengali: Dataset creation and multilingual LLM integration, 2025.

[8] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, 2017. arXiv:1706.04599.

[9] Hazem Ibrahim and Yasir Zaki. Evaluating decision models for text annotation in computational social science, 2026.

[10] Khondoker Ittehadul Islam, Md Saiful Islam, Sudipta Kar, and Mohammad Ruhul Amin. SentNoB: A dataset for analysing sentiment on noisy Bangla texts. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, 2021. ACL Anthology 2021.findings-emnlp.278.

[11] Md Mahir Jawad, Galib Mahmud Jim, Rafid Ahmed, Mir Sazzat Hossain, Md Fahim, and Md Farhad Alam Bhuiyan. 5-dialects-BN: Unmasking the impact of transliteration on Bangla dialectal LLMs, 2026.

[12] Manurag Khullar, Utkarsh Desai, Poorva Malviya, Aman Dalmia, and Zheyuan Ryan Shi. Script gap: Evaluating LLM triage on Indian languages in native vs romanized scripts in a real world setting, 2025.

[13] Ran Li and Lei Chen. Readout stability in prefill-only decision models: Zero-label prediction and inference-time compute allocation, 2026.

[14] Xunzhuo Liu, Huamin Chen, Samzong Lu, Yossi Ovadia, et al. vLLM semantic router: Signal driven decision routing for mixture-of-modality models, 2026.

[15] Marc Marone, Orion Weller, William Fleshman, Eugene Yang, Dawn Lawrie, and Benjamin Van Durme. mmBERT: A modern multilingual encoder with annealed language learning, 2025.

[16] Gowthamkumar Nandakishore. Laya as a typed probabilistic assessor: An independent reproduction and a preregistered study of calibration and selective escalation, 2026.

[17] Rahul Sharma, Andrew B. Duncan, Gaétan Marceau Caron, and Sebastian J. Vollmer. TypedBench: A benchmark for calibration, framing sensitivity, and cost in system one decision models, 2026.

[18] Yu Sun, Junhao Xu, Jiajia Shi, et al. Type-safe is not error-free: Typed decision models follow the option name, not the definition bound to it, 2026.

[19] Zexiao Wang, Zihao Zhang, Xudong Wang, Pan Wang, et al. Chinese-Jev: Bringing system one model to Chinese-language tasks, 2026.

[20] Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations*, 2024. arXiv:2309.03882.

A Additional tables

	AI Banglish − rule-based				AI Banglish − Bengali			
Model	MASSIVE	SentNoB	XNLI-bn	task mean	MASSIVE	SentNoB	XNLI-bn	task mean
Jev	+13.6 [+9.7, +17.4]	+0.0 [-4.9, +4.6]	+8.6 [+0.0, +17.2]	+7.4	-12.8 [-16.6, -8.7]	-8.9 [-13.9, -3.7]	-10.3 [-20.7, +1.7]	-10.7
d3-flash (4-bit)†	+20.4 [+16.0, +24.7]	+5.8 [+1.8, +10.2]	-1.7 [-15.5, +12.1]	+8.2	-19.4 [-24.1, -15.2]	-10.2 [-14.8, -5.5]	-17.2 [-32.8, -3.4]	-15.6
Gemma-4-E4B	+15.6 [+12.1, +19.6]	+6.2 [+1.8, +10.2]	+5.2 [-1.7, +13.8]	+9.0	-10.9 [-14.6, -7.1]	-4.6 [-8.9, +0.0]	+0.0 [-10.3, +8.6]	-5.2
d3-mini	+16.4 [+12.3, +20.4]	+2.8 [-1.2, +6.5]	+6.9 [-1.7, +17.2]	+8.7	-24.5 [-29.4, -19.6]	-10.8 [-15.4, -5.5]	-8.6 [-24.2, +8.6]	-14.6
d3-nano	+20.2 [+16.0, +24.5]	+6.5 [+1.8, +10.8]	+5.2 [-6.9, +19.0]	+10.6	-24.7 [-30.2, -19.4]	-11.7 [-17.5, -5.8]	-5.2 [-20.7, +10.3]	-13.9
Kev-4B	+17.8 [+14.0, +21.9]	+1.5 [-2.2, +5.5]	+1.7 [-6.9, +10.3]	+7.0	-20.4 [-25.7, -15.8]	-7.1 [-12.6, -1.5]	-13.8 [-25.9, -1.7]	-13.8
d3-lite	+19.8 [+16.0, +24.1]	+7.1 [+2.8, +12.0]	+5.2 [-8.6, +17.2]	+10.7	-25.5 [-30.8, -21.2]	-14.8 [-21.5, -8.0]	-15.5 [-32.8, +1.7]	-18.6
laya-bangla	+18.2 [+14.6, +22.3]	+0.9 [-2.5, +4.3]	+0.0 [-15.5, +15.5]	+6.4	-34.0 [-38.9, -28.9]	-12.3 [-18.2, -6.8]	+1.7 [-17.2, +20.7]	-14.9
d3-edge	+25.5 [+21.7, +29.2]	+5.2 [+0.6, +9.8]	+12.1 [+1.7, +24.1]	+14.3	+6.9 [+2.0, +11.9]	-6.2 [-12.0, -0.3]	-8.6 [-25.9, +8.6]	-2.6
Kev-0.8B	+19.2 [+15.4, +23.3]	+1.8 [-2.5, +6.2]	-1.7 [-13.8, +12.1]	+6.5	-7.5 [-11.7, -2.6]	-1.2 [-6.8, +4.3]	-8.6 [-20.7, +1.8]	-5.8
Laya-multilingual	+14.0 [+10.7, +17.4]	+1.2 [-0.6, +3.4]	+8.6 [-5.2, +22.4]	+7.9	-1.6 [-5.9, +2.6]	-1.2 [-4.3, +1.8]	-10.3 [-27.6, +8.6]	-4.4
Julia-1	–	-7.4 [-13.2, -1.8]	+3.4 [-12.1, +19.0]	–	–	-4.6 [-11.7, +2.2]	-3.4 [-17.2, +10.4]	–

Table 8: The two differences of Table 6 for each task, in points with 95% bootstrap intervals, and the unweighted mean of the three task differences. XNLI-bn has only 58 pairs. Julia-1 cannot take MASSIVE.

Model	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.918 [0.897, 0.936]	0.897 [0.853, 0.936]	0.751 [0.729, 0.772]	0.757 [0.735, 0.779]	0.623 [0.596, 0.647]
Gemma-4-E4B	0.773 [0.743, 0.801]	0.853 [0.804, 0.897]	0.631 [0.606, 0.655]	0.689 [0.667, 0.713]	0.590 [0.566, 0.615]
d3-mini	0.727 [0.697, 0.758]	0.838 [0.789, 0.887]	0.729 [0.705, 0.752]	0.749 [0.728, 0.771]	0.579 [0.555, 0.604]
d3-nano	0.702 [0.671, 0.733]	0.848 [0.799, 0.897]	0.746 [0.725, 0.768]	0.705 [0.681, 0.730]	0.592 [0.568, 0.614]
Kev-4B	0.561 [0.528, 0.592]	0.779 [0.721, 0.833]	0.624 [0.603, 0.648]	0.575 [0.551, 0.600]	0.536 [0.511, 0.556]
d3-lite	0.577 [0.542, 0.609]	0.760 [0.701, 0.819]	0.704 [0.681, 0.726]	0.624 [0.600, 0.647]	0.535 [0.511, 0.558]
laya-bangla	0.313 [0.283, 0.344]	0.750 [0.686, 0.814]	0.625 [0.599, 0.650]	0.575 [0.547, 0.602]	0.464 [0.437, 0.488]
d3-edge	0.409 [0.377, 0.441]	0.475 [0.407, 0.544]	0.523 [0.497, 0.548]	0.243 [0.223, 0.264]	0.472 [0.451, 0.498]
Kev-0.8B	0.374 [0.341, 0.407]	0.632 [0.564, 0.701]	0.469 [0.445, 0.491]	0.372 [0.347, 0.397]	0.359 [0.337, 0.380]
Laya-multilingual	0.287 [0.259, 0.314]	0.588 [0.515, 0.657]	0.635 [0.612, 0.660]	0.179 [0.161, 0.199]	0.263 [0.244, 0.283]
Julia-1	0.287 [0.258, 0.316]	0.520 [0.456, 0.588]	0.382 [0.359, 0.406]	–	0.403 [0.379, 0.424]

Table 9: Accuracy on Bengali script with 95% cluster-bootstrap intervals (SentNoB: macro-F1).

Model	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.9 / 93%	1.0 / 97%	0.5 / 97%	1.7 / 89%	0.9 / 95%
Gemma-4-E4B	1.4 / 74%	0.5 / 93%	3.3 / 86%	4.3 / 70%	0.1 / 94%
d3-nano	1.1 / 82%	0.5 / 96%	0.7 / 96%	1.1 / 85%	0.4 / 95%
d3-lite	1.2 / 82%	1.0 / 96%	0.7 / 96%	0.5 / 80%	1.2 / 92%
laya-bangla	2.1 / 28%	2.0 / 94%	1.2 / 89%	0.3 / 66%	2.8 / 88%
d3-edge	0.3 / 75%	4.4 / 83%	1.9 / 92%	3.3 / 49%	1.4 / 90%
Laya-multilingual	1.1 / 15%	1.0 / 84%	0.1 / 91%	8.7 / 29%	4.6 / 80%
Julia-1	1.8 / 31%	2.9 / 74%	1.9 / 24%	–	10.8 / 45%

Table 10: Spread of accuracy across the three option rotations (points) / share of items answered identically under all three.

	XNLI-bn		MASSIVE		SentNoB	
Model	Bengali	Rule	Bengali	Rule	Bengali	Rule
Jev	0.088	0.197	0.087	0.081	0.202	0.161
Gemma-4-E4B	0.327	0.402	0.249	0.397	0.333	0.374
d3-mini	0.049	0.038	0.054	0.104	0.120	0.094
d3-nano	0.069	0.071	0.073	0.129	0.107	0.165
Kev-4B	0.238	0.316	0.270	0.101	0.170	0.205
d3-lite	0.059	0.057	0.048	0.272	0.120	0.187
laya-bangla	0.203	0.286	0.071	0.138	0.339	0.568
d3-edge	0.178	0.066	0.223	0.423	0.120	0.117
Kev-0.8B	0.173	0.131	0.166	0.053	0.192	0.247
Laya-multilingual	0.276	0.469	0.236	0.468	0.530	0.654
Julia-1	0.420	0.532	–	–	0.444	0.456

Table 11: ECE (15 bins, no temperature scaling) within each task on the full test sets, for Bengali script and rule-based romanization. The mean of these per-task values over the three tasks is the primary ECE of Table 7. d3-flash is omitted (subset).

Model	Bengali	Rule-based	AI-generated Banglish
Jev	0.117	0.109	0.101
Gemma-4-E4B	0.291	0.415	0.335
d3-mini	0.100	0.106	0.106
d3-nano	0.115	0.148	0.128
Kev-4B	0.159	0.131	0.180
d3-lite	0.113	0.251	0.192
laya-bangla	0.142	0.299	0.257
d3-edge	0.204	0.331	0.209
Kev-0.8B	0.150	0.148	0.120
Laya-multilingual	0.371	0.536	0.487
Julia-1	0.433	0.395	0.444

Table 12: Raw ECE (15 bins, no temperature scaling) on the 877 AI-generated Banglish items, pooled over MASSIVE, SentNoB and XNLI-bn pairs, in three conditions of the same items. d3-flash is omitted (subset).

	Brier score		Confidence − accuracy	
Model	Bengali	Rule-based	Bengali	Rule-based
Jev	0.418	0.617	+0.122	+0.144
Gemma-4-E4B	0.637	0.849	+0.303	+0.391
d3-mini	0.432	0.649	+0.069	+0.073
d3-nano	0.429	0.731	+0.081	+0.121
Kev-4B	0.616	0.802	+0.045	+0.141
d3-lite	0.508	0.806	+0.073	+0.166
laya-bangla	0.650	0.983	+0.159	+0.331
d3-edge	0.744	0.852	+0.173	+0.201
Kev-0.8B	0.734	0.817	+0.066	+0.136
Laya-multilingual	0.909	1.216	+0.347	+0.530
Julia-1	0.972	1.080	+0.431	+0.494

Table 13: Brier score and confidence minus accuracy on Bengali script and rule-based romanization (test sets, mean over XNLI-bn, MASSIVE and SentNoB, no temperature scaling). Positive values mean overconfident.